

Lebanese University – Faculty of Engineering

Subject: Sensors

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**Sensors project proposal
(Greenhouse)**

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1. Scenario Description

Traditional greenhouse management relies heavily on manual labor and periodic human observation, which introduces significant risks: delayed response to thermal stress or soil dehydration, resource inefficiency caused by over-irrigation, and a lack of automated entry logging or tamper alarms that leaves crops vulnerable to unauthorized access. This project proposes an autonomous, closed-loop smart greenhouse built around an Arduino Mega 2560 microcontroller. The system will continuously monitor temperature, humidity, soil moisture, water reservoir level, ambient light, sound, and proximity, and will respond automatically by driving an irrigation pump, a ventilation fan, grow-light indicators, and an audible alarm. Access to the system will be gated by an RFID card reader so that only authorized users can arm or interact with the greenhouse controls.

2. System Architecture

The system follows a classic Sensing $\rightarrow$ Processing $\rightarrow$ Actuation control loop. The Arduino Mega 2560 is the central controller. Sensors connect through analog inputs (A0–A2, A15), digital I/O (D2–D53), the SPI bus (RC522 RFID reader), and a hardware interrupt pin (D2) for the water-flow sensor. A 12V external adapter powers the irrigation pump through a 2-channel relay module, electrically isolating the pump circuit from the microcontroller logic. A 16x2 HD44780 LCD will display six pages of live data, cycled with a single push button. All actuator logic will be written using millis()-based non-blocking timers so the system stays responsive at all times.

Component	Arduino Pin(s)	Function
LCD 16x2	D22– D23-D24-D25-D26_D27	Displays sensor data, cycles through 6 pages
DHT11	D32	Ambient temperature & humidity
RC522 RFID	D9, D6, D50–D52	Access control / system arming
HC-SR04	D28 , D29	Proximity / intrusion detection
Water level sensor	A0	Reservoir level monitoring
Soil moisture sensor	A1	Irrigation trigger
LDR	A2	Ambient light for grow-LED control
Sound sensor	A15 (AO), A14 (DO)	Acoustic anomaly detection
flow sensor	D2	Irrigation volume tracking
Soil relay	D34	Automated irrigation
DC fan	D3	Cooling / ventilation
Buzzer	D53	Audible alarm
Push button	D33	LCD page navigation

Water LED	D48 D44 D46	lights D48/D44/D46 for low/medium/full
RFID Green LED	D36	Accept access
RFID Red LED	D40	Access denied
LDR LED	D38 D42	Turn on when it become dark

3. Bill of Materials:

Item	Qty	price
Ultimate Mega Kit Arduino	1	60\$
12mm LDR	1	0.5\$
YF-S201 water flow sensor	1	8\$
2-channel relay module	1	2\$
12V DC water pump	1	5\$
12V DC adapter	1	5\$
Total		80.5\$